

IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 02 ASSIGNMENT REPORT

FLSM / VLSM IP Addressing and VLAN Configuration — Practical Implementation

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1. Introduction

This report documents the practical implementation of an IP addressing and VLAN segmentation exercise carried out during Week 02 of the IT-Simplera Network Administration Internship. The task required designing a subnetting scheme for four departments with differing host requirements, then implementing and verifying that scheme across two separate lab environments: Cisco Packet Tracer and GNS3.

Because the four VLANs required a different number of usable host addresses, a single fixed mask could not be applied efficiently across all of them. Variable Length Subnet Masking (VLSM) was therefore used to size each subnet to its actual requirement, minimising wasted address space while keeping every subnet within the single major network 192.168.90.0/24.

2. Addressing Problem Statement

Design an efficient IP addressing scheme for the major network 192.168.90.0/24, to be divided across four VLANs with the following host requirements. Each VLAN must receive the smallest subnet that satisfies its requirement, and no address space may overlap.

VLAN	Department / Segment	Hosts Required	Colour Code
10	Sales / Pink Zone	50	Pink
20	Engineering / Green Zone	30	Green
30	Management / Blue Zone	10	Blue
40	IT / Teal Zone	10	Teal

Total hosts requested: 50 + 30 + 10 + 10 = 100 addresses, drawn from a /24 block of 254 usable addresses.

3. Subnetting Solution (VLSM Breakdown)

Subnets were allocated largest-requirement-first to avoid fragmentation. The resulting breakdown is shown below:

VLAN	Network Address	Subnet Mask	CIDR	Usable Host Range	Broadcast
10	192.168.90.0	255.255.255.192	/26	192.168.90.1 - .62	192.168.90.63
20	192.168.90.64	255.255.255.224	/27	192.168.90.65 - .94	192.168.90.95
30	192.168.90.96	255.255.255.240	/28	192.168.90.97 - .110	192.168.90.111
40	192.168.90.112	255.255.255.240	/28	192.168.90.113 - .126	192.168.90.127

- VLAN 10 (50 hosts) → borrowed 2 bits → /26 → 62 usable hosts
- VLAN 20 (30 hosts) → borrowed 3 bits → /27 → 30 usable hosts
- VLAN 30 (10 hosts) → borrowed 4 bits → /28 → 14 usable hosts

- VLAN 40 (10 hosts) → borrowed 4 bits → /28 → 14 usable hosts

4. Topology — Cisco Packet Tracer Implementation

The topology below shows two access-layer switches, each carrying all four VLANs, interconnected by an 802.1Q trunk link. End devices in each colour-coded zone correspond to the VLAN allocation defined above.

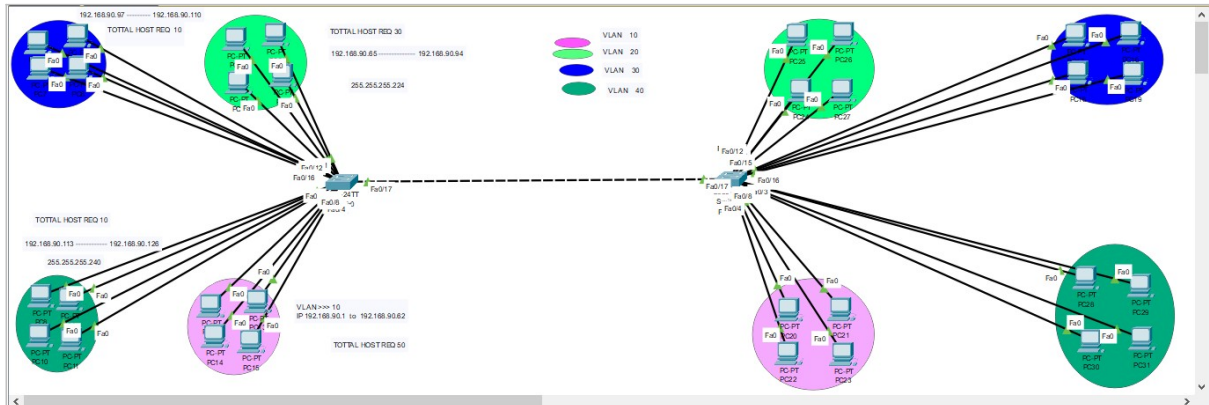


Figure 1 — Packet Tracer topology showing VLAN 10/20/30/40 zones and inter-switch trunk.

5. Configuration — Cisco Packet Tracer

5.1 Trunk Configuration Verification

The trunk link between the two switches was verified using "show interface trunk", confirming 802.1Q encapsulation and VLANs 10, 20, 30 and 40 active in the management domain:

Switch(config)#do sho int trunk				
Port	Mode	Encapsulation	Status	Native vlan
Fa0/17	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Fa0/17	1-1005			
Port	Vlans allowed and active in management domain			
Fa0/17	1,10,20,30,40			
Port	Vlans in spanning tree forwarding state and not pruned			
Fa0/17	none			

Figure 2 — show interface trunk output on Fa0/17 (Packet Tracer).

5.2 End-Device IP Configuration

Representative end devices were statically addressed according to their VLAN's allocated subnet, and configured with the correct subnet mask:

Device	IP Address	Subnet Mask	VLAN
PC3	192.168.90.66	255.255.255.224	20
PC4	192.168.90.98	255.255.255.240	30

Device	IP Address	Subnet Mask	VLAN
PC8	192.168.90.113	255.255.255.240	40
PC13	192.168.90.1	255.255.255.0	10 (test host)

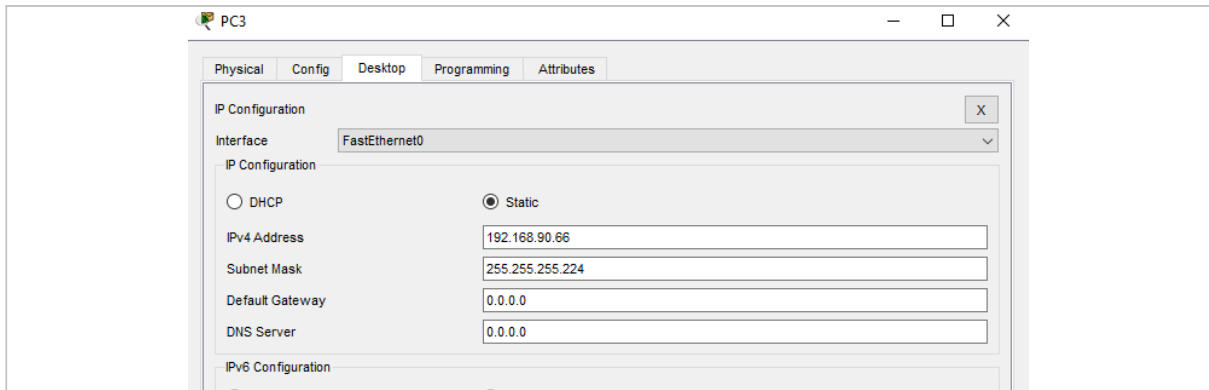


Figure 3 — PC3 static IP configuration (VLAN 20).

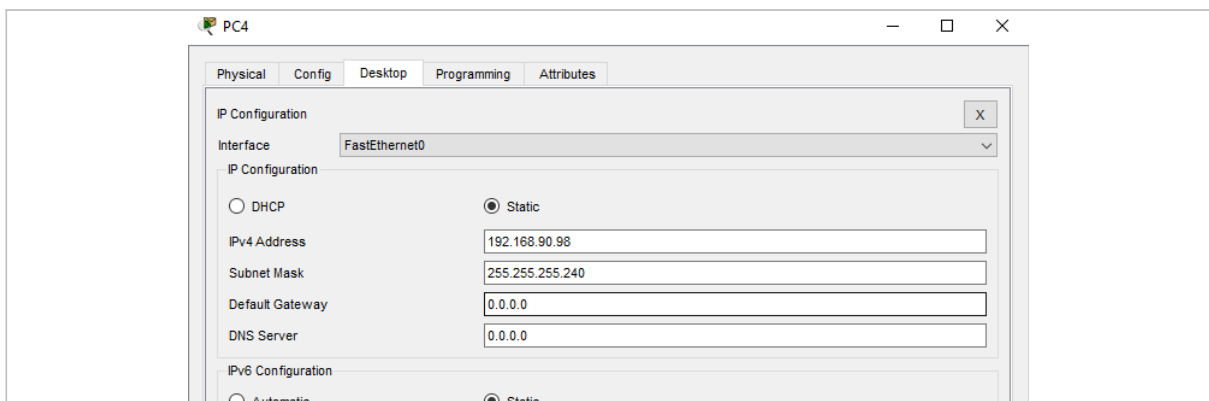


Figure 4 — PC4 static IP configuration (VLAN 30).

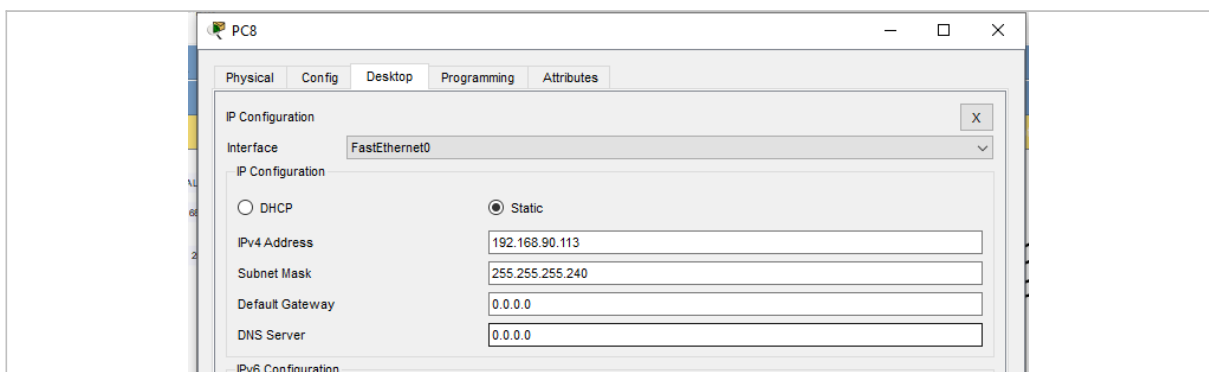


Figure 5 — PC8 static IP configuration (VLAN 40).

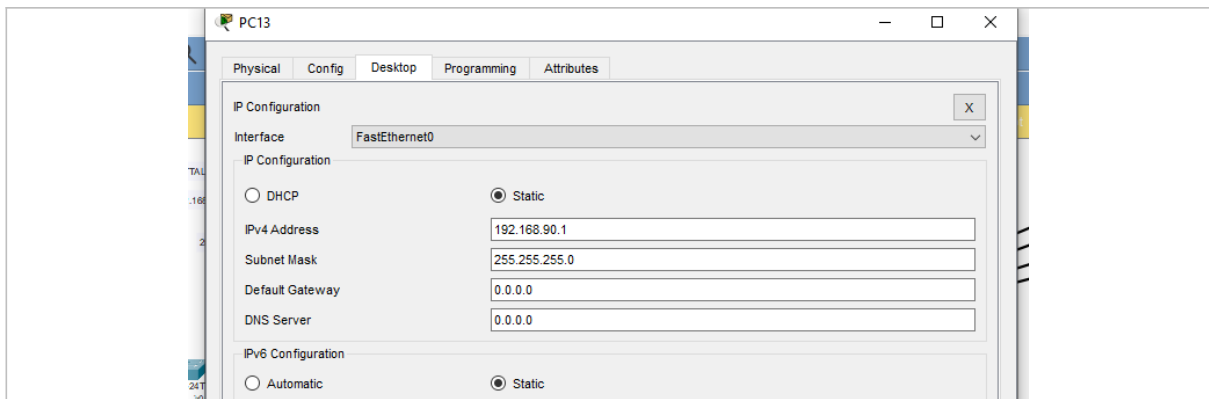


Figure 6 — PC13 static IP configuration (VLAN 10 test host).

6. Topology — GNS3 Implementation

The same addressing design was rebuilt in GNS3 using VPCS end nodes across two switches (Switch2 and Switch3) connected by a trunk, reinforcing the configuration in a second, more CLI-driven environment:

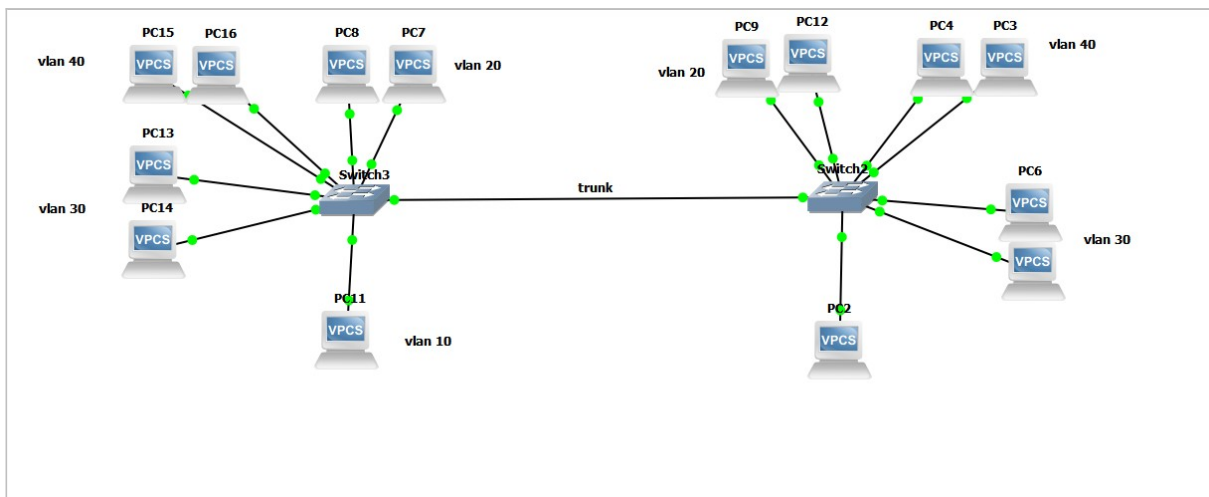


Figure 7 — GNS3 topology: Switch2 and Switch3 with VLAN 10/20/30/40 VPCS nodes and inter-switch trunk.

7. Configuration — GNS3

7.1 Trunk Configuration Verification

The GNS3 trunk interface (Et1/0) was verified in the same way, confirming 802.1Q trunking with VLANs 10, 20, 30 and 40 active and forwarding:

```

Switch#
Switch#
Switch#sho int trunk

Port      Mode      Encapsulation  Status      Native vlan
Et1/0     on        802.1q         trunking    1

Port      Vlans allowed on trunk
Et1/0     1-4094

Port      Vlans allowed and active in management domain
Et1/0     1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Et1/0     1,10,20,30,40
Switch#

```

Figure 8 — show interface trunk output on Et1/0 (GNS3).

7.2 VPCS End-Device Verification and Ping Tests

Connectivity within each VLAN was confirmed using VPCS "show ip" and inter-host ping tests, all returning 0% packet loss:

```

VPCS> sho ip

NAME       : VPCS[1]
IP/MASK    : 192.168.90.2/26
GATEWAY    : 255.255.255.192
DNS        :
MAC        : 00:50:79:66:68:27
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.90.1

84 bytes from 192.168.90.1 icmp_seq=1 ttl=64 time=0.546 ms
84 bytes from 192.168.90.1 icmp_seq=2 ttl=64 time=3.876 ms
84 bytes from 192.168.90.1 icmp_seq=3 ttl=64 time=3.105 ms
84 bytes from 192.168.90.1 icmp_seq=4 ttl=64 time=1.156 ms

```

Figure 9 — VPCS[1] at 192.168.90.2/26 (VLAN 10) pinging gateway 192.168.90.1.

```

VPCS> sho ip

NAME       : VPCS[1]
IP/MASK    : 192.168.90.120/28
GATEWAY    : 255.255.255.240
DNS        :
MAC        : 00:50:79:66:68:2c
LPORT     : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.90.114

84 bytes from 192.168.90.114 icmp_seq=1 ttl=64 time=2.244 ms
84 bytes from 192.168.90.114 icmp_seq=2 ttl=64 time=1.682 ms
84 bytes from 192.168.90.114 icmp_seq=3 ttl=64 time=5.003 ms
84 bytes from 192.168.90.114 icmp_seq=4 ttl=64 time=1.866 ms
84 bytes from 192.168.90.114 icmp_seq=5 ttl=64 time=1.973 ms

```

Figure 10 — VPCS[1] at 192.168.90.120/28 (VLAN 40) pinging 192.168.90.114.

```
VPCS> sho ip

NAME       : VPCS[1]
IP/MASK    : 192.168.90.100/28
GATEWAY    : 255.255.255.240
DNS        :
MAC        : 00:50:79:66:68:07
LPORT      : 20000
RHOST:PORT : 127.0.0.1:30000
MTU        : 1500

VPCS> ping 192.168.90.97

84 bytes from 192.168.90.97 icmp_seq=1 ttl=64 time=1.405 ms
84 bytes from 192.168.90.97 icmp_seq=2 ttl=64 time=4.112 ms
84 bytes from 192.168.90.97 icmp_seq=3 ttl=64 time=2.048 ms
84 bytes from 192.168.90.97 icmp_seq=4 ttl=64 time=2.654 ms
84 bytes from 192.168.90.97 icmp_seq=5 ttl=64 time=1.793 ms
```

Figure 11 — VPCS[1] at 192.168.90.100/28 (VLAN 30) pinging 192.168.90.97.

8. Key Learning Outcomes

- VLSM allows subnet size to match actual host requirements, reducing wasted address space compared to FLISM.
- Trunk links must explicitly allow every VLAN required in the management domain, or traffic for that VLAN will be pruned.
- Verification commands (show interface trunk, show ip, ping) are essential to prove a design works, not just that it was configured.
- Rebuilding the same topology in a second tool (GNS3) after Packet Tracer reinforced CLI fluency and reduced platform-specific assumptions.

Grateful to my supervisor, Jawad Qayum, for the guidance throughout this exercise.